

MERN Stack Internship Live Project

Project Title

Employee Attendance & Task Management System

Project Overview

Build a web application where Admin can manage employees, assign tasks, track attendance, and generate reports.

Module 1: Authentication & Authorization

Task 1: User Authentication

Features

- User Registration
- User Login
- Forgot Password
- Reset Password
- JWT Authentication
- Logout

Requirements

- Password hashing using bcrypt
- JWT token generation
- Access token validation middleware
- Role-based access control

Roles

- Admin
 - Employee
-

Module 2: Employee Management

Task 2: Employee CRUD

Features

- Add Employee
- Edit Employee
- Delete Employee
- View Employee Details
- Employee Listing

Fields

- Employee ID
- Full Name
- Email
- Mobile Number
- Department
- Designation
- Joining Date
- Profile Image

Requirements

- Server-side validation
 - Pagination
 - Search
 - Sorting
-

Module 3: Attendance Management

Task 3: Attendance System

Features

- Mark Attendance
- Check-In
- Check-Out
- Attendance History

Attendance Status

- Present
- Absent
- Half Day
- Leave

Requirements

- One attendance per day
 - Attendance report API
 - Monthly attendance summary
-

Module 4: Task Management

Task 4: Task Assignment

Features

- Create Task
- Assign Employee
- Update Status
- Add Comments
- Due Date Management

Status

- Pending
- In Progress
- Completed
- Rejected

Requirements

- Task history tracking
 - Activity logs
-

Module 5: Leave Management

Task 5: Leave Module

Features

- Apply Leave
- Approve Leave
- Reject Leave
- Leave History

Leave Types

- Casual Leave
- Sick Leave
- Emergency Leave

Requirements

- Leave balance calculation
 - Leave approval workflow
-

Module 6: Dashboard

Task 6: Dashboard Analytics

Admin Dashboard

Display:

- Total Employees
- Present Today
- Absent Today
- Pending Tasks
- Completed Tasks
- Leave Requests

Charts

- Attendance Chart
 - Task Status Chart
 - Employee Growth Chart
-

Module 7: File Upload

Task 7: Document Management

Upload Documents

- Resume
- Aadhaar
- PAN Card
- Certificates

Requirements

- Multer Integration
 - File Type Validation
 - File Size Validation
-

Module 8: Notification System

Task 8: Real-Time Notifications

Features

- New Task Assigned
- Leave Approved
- Leave Rejected
- Task Completed

Technology

- Socket.io
-

Module 9: Audit Logs

Task 9: Activity Tracking

Track:

- Login
 - Logout
 - Employee Added
 - Employee Updated
 - Attendance Marked
 - Leave Approved
-

Module 10: Reports

Task 10: Report Module

Generate:

- Attendance Report
- Employee Report
- Leave Report
- Task Report

Export Formats

- Excel
 - PDF
-

Technical Requirements

Backend (Node.js + Express)

Folder Structure

```
src/  
├── config/  
├── controllers/  
├── middleware/  
├── models/  
├── routes/  
├── services/  
└── validations/
```

```
|— utils/  
|— uploads/  
|— sockets/  
|— app.js
```

Database

MongoDB Collections

users
employees
attendance
tasks
leaves
documents
notifications
activity_logs

Frontend (React.js)

Pages

Login
Register
Dashboard
Employees
Attendance
Tasks
Leaves
Reports
Profile
Settings

Requirements

- React Router
- Axios
- Redux Toolkit / Context API
- Protected Routes
- Form Validation
- Loading State
- Error Handling
- Responsive UI